FedShield-IDS: A Federated, Adversarially-Robust Intrusion Detection System for Heterogeneous IoT Networks

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***Abstract*—Deep learning models for network intrusion detection perform well in offline benchmarks but face severe operational hurdles when deployed across decentralized Internet of Things (IoT) networks. First, privacy regulations and bandwidth limits prohibit streaming raw packet telemetry to a central server, mandating decentralized learning. Second, decentralized edge models remain vulnerable to two distinct attacks: inference-time packet evasion and training-time Byzantine poisoning by compromised edge devices. Existing Byzantine-robust aggregation rules assume client traffic is identically distributed; in real networks with non-IID traffic, they frequently misclassify benign, specialized edge traffic as adversarial poisoning.**

We present FedShield-IDS, an adversarially robust federated intrusion detection architecture designed for heterogeneous edge networks. At the edge, clients harden local 1D-CNNs using Projected Gradient Descent (PGD) adversarial training, enforcing a feature mutability mask so simulated perturbations respect physical transport protocol constraints. At the central coordinator, incoming updates pass through a two-tier filter combining layer-wise Frobenius dispersion scoring with cosine directional consensus. This allows the server to detect and discard poisoned model updates while accommodating legitimate non-IID variance from honest edge nodes. Evaluated on all 286,467 real network flows from the CIC-IDS2017 benchmark under Dirichlet non-IID partitioning ($\beta = 0.5$), FedShield-IDS achieves 99.58% clean classification accuracy, reduces black-box evasion attack success rates from 100.00% to 0.95% (a 99.05 percentage point defensive gain), and blocks 100% of Byzantine poisoned updates while maintaining an edge inference latency below 0.12 ms per flow.

***Index Terms*—Intrusion Detection System, Federated Learning, Adversarial Robustness, IoT Security, Byzantine Resilience, Deep Learning**

I. Introduction

Edge Internet of Things (IoT) networks present a difficult security problem: resource-constrained appliances with diverse firmware are continuously exposed to automated cyber threats, ranging from distributed denial-of-service botnets to endpoint exploits. Deep learning has emerged as an effective tool for Network Intrusion Detection Systems (NIDS), with one-dimensional convolutional neural networks (1D-CNNs) [1] and hybrid federated learning architectures [2] achieving detection accuracies above 98% in offline benchmarks.

However, deploying deep NIDS across operational IoT infrastructures introduces two practical challenges:

1) Telemetry Privacy and Distributed Governance: Classical NIDS architectures require edge devices to stream raw packet captures or flow metadata to a central Security Operations Center (SOC). In enterprise and healthcare deployments, centralizing raw flows violates privacy regulations (such as GDPR and HIPAA) and exposes sensitive internal network structure. Federated Learning (FL) [3] offers an alternative by training models locally: edge nodes compute gradient updates on private flow data and send only model parameter deltas to a central coordinator.

2) Dual-Vector Adversarial Vulnerability: While federated learning keeps raw data private, decentralizing optimization removes the traditional network perimeter and exposes the system to two distinct adversarial threats:

- *Inference-Time Evasion:* Deep learning classifiers are sensitive to small input perturbations [4]. Attackers can craft subtle flow modifications, such as padding packet sizes or adjusting timing, that evade detection without breaking network protocols [5].
- *Training-Time Byzantine Model Poisoning:* Central aggregation is vulnerable to compromised edge clients [6]. Hijacked IoT nodes can send poisoned parameter updates designed to shift decision boundaries, creating backdoors for specific attacks or disrupting global model convergence.

The Non-IID Defense Dilemma: Existing research generally examines these issues in isolation. Federated intrusion detection proposals [7], [8] address Byzantine poisoning but assume cooperative, identically distributed clients, whereas adversarial defense studies assume centralized environments [9]. Combining these defenses naively introduces an unresolved tension with statistical data heterogeneity (non-IID traffic). In real IoT networks, different device types observe divergent traffic patterns, such as continuous video streams versus periodic sensor telemetry. Conventional Byzantine aggregation rules, including Krum, Trimmed Mean, and coordinate medians, assume client data is identically distributed; under non-IID conditions, they routinely misidentify benign client variance as malicious poisoning. At the same time, client-side

adversarial training increases gradient variance across nodes, causing honest updates to be discarded by naive server filters.

To address these challenges together, an operational IoT NIDS must provide three guarantees: raw-data localization for privacy, edge robustness against evasion attacks, and server-side Byzantine resilience that distinguishes malicious manipulation from non-IID statistical heterogeneity.

A. Contributions

In this paper, we propose **FedShield-IDS**, a cohesive federated intrusion detection framework engineered for heterogeneous IoT ecosystems. FedShield-IDS harmonizes edge-level adversarial regularization with coordinator-level normalized directional filtering. Our core contributions are:

- **Integrated Defense-in-Depth Architecture:** We design an end-to-end framework combining client-side Projected Gradient Descent (PGD) adversarial training on compact 1D-CNNs (186K parameters) with a two-tier server-side aggregation protocol combining layer-wise Frobenius normalization and cosine directional consensus.
- **Heterogeneity-Preserving Byzantine Resilience:** We formulate a kernel-normalized anomaly scoring metric that scales parameter updates by layer-wise dispersion relative to the median update trajectory, preventing benign non-IID clients from being falsely rejected while systematically isolating Byzantine corruptions.
- **Formal Dual-Threat Characterization:** We formalize a comprehensive threat model capturing simultaneous inference evasion under strict ℓ_∞-bounded black-box query budgets and up to 20% Byzantine poisoning participants executing targeted model subversion across non-IID Dirichlet traffic partitions ($\beta = 0.5$).
- **Rigorous Empirical Validation on Real Hardware:** We benchmark FedShield-IDS on all 286,467 real network flows from the CIC-IDS2017 dataset. FedShield-IDS attains 99.58% clean classification accuracy under Dirichlet non-IID partitioning ($\beta = 0.5$), suppresses black-box evasion Attack Success Rate (ASR) from 100.00% down to 0.95% (a 99.05 percentage point defensive margin), blocks 100% of Byzantine poisoned updates across communication rounds, and sustains line-rate flow inspection with an inference latency below 0.12 ms per flow.

B. Paper Organization

The remainder of this paper is organized as follows: Section II synthesizes related work. Section III formalizes the system model and dual-threat formulation. Section IV details the FedShield-IDS architecture, including Algorithms 1 and 2. Section V presents empirical evaluation results, baseline comparisons, and ablation studies. Section VI discusses operational trade-offs, and Section VII concludes the paper.

II. Related Work

A. Deep Learning for Network Intrusion Detection

Sommer and Paxson [11] identified the core mismatch between anomaly detection classifiers and production network environments: benign traffic is so diverse that any model trained offline accumulates false alarms at operational scale. Kushwaha et al. [12] surveyed intelligent IDS across IoT ecosystems and confirmed that deep feature extraction reduces false alarm rates relative to classical rule-based detectors, though model complexity grows with device heterogeneity. Hnamte and Hussein [1] showed that a one-dimensional convolutional neural network (1D-CNN) applied directly to raw flow feature vectors achieves sub-0.10 ms per-sample inference on IoT gateway hardware by exploiting local spatial correlations across feature channels without attention overhead. Omotosho et al. [13] evaluated federated intrusion detection on MQTT protocol traffic, finding that distributing training across brokers preserves per-protocol detection accuracy while keeping raw payloads off the central server. Jeribi et al. [10] designed a hierarchical two-stage detector for IoT gateways that retains classification accuracy under partial data corruption, though it does not defend against gradient-based evasion during inference. Ohtani et al. [14] demonstrated that autonomously extracted anomaly signals allow a federated model to flag zero-day traffic without labeled attack traces, at the cost of higher false positive rates on rare-but-benign flows. Chowdhury et al. [2] evaluated FLEX-IDS on CIC-IDS2017 under adversarial edge conditions and found that explainability-guided feature selection recovers detection accuracy degraded by feature-level perturbations, but the explainability pipeline adds inference latency that exceeds line-rate constraints on constrained gateways. All of these architectures assume that input telemetry arrives unperturbed at inference time. None of them harden the model against an adversary who crafts traffic specifically to cross the decision boundary.

B. Federated Learning in Edge Security

McMahan et al. [3] introduced FedAvg, in which each client performs several steps of stochastic gradient descent on local data before the server averages the resulting weight deltas. Standard FedAvg aggregation has no mechanism to distinguish a Byzantine poisoned update from a legitimate update produced by a client observing skewed class proportions. Blanchard et al. [15] proposed Krum, which selects the update geometrically closest to its neighbors, but this criterion breaks down when honest clients hold non-IID traffic distributions because legitimate updates diverge in parameter space by design. Yin et al. [16] showed that coordinate-wise median provides provable Byzantine tolerance under IID assumptions, but the same non-IID skew causes the median to suppress honest minority updates. Bagdasaryan et al. [6] demonstrated that a single compromised client can steer the global model using weight scaling, bypassing distance-based filters entirely. Gad et al. [17] addressed this by designing a robust FL aggregation pipeline for IoT non-IID settings, showing that careful client selection reduces poisoning impact, though convergence slows significantly when honest clients are excluded conservatively. Sankeawthong et al. [7] built TrustFed-IDS, a trust-scored MLP-LSTM aggregation scheme that down-weights low-reputation clients and achieves Byzantine re-

TABLE I
SYSTEMATIC COMPARISON WITH RELATED IDS ARCHITECTURES

System / Architecture	Decentralized (FL)	Evasion-Robust	Poisoning-Robust	Non-IID Resilient	Line-Rate Edge Latency
Lightweight CNN IDS [1]	×	×	×	Partial	✓(<0.10 ms)
FLEX-IDS [2]	✓	Partial	Partial	✓	Partial (≈0.60 ms)
TrustFed-IDS [7]	✓	×	✓	Partial	✓(<0.20 ms)
EdgeGuard-IoT [10]	×	Partial	×	✓	✓(<0.15 ms)
FedShield-IDS (This Work)	✓	✓	✓	✓	**✓(<0.12 ms)**

silience, but evaluated it under IID partitions where trust scores stabilize quickly and non-IID divergence never challenges the scoring model. Sultan et al. [18] combined differential privacy with selective aggregation in FLSDF to protect edge IoT fusion against inference attacks, reducing leakage at the cost of model accuracy under high privacy budgets. Masood et al. [19] integrated reinforcement-based client scheduling into federated detection for IoT-UAV environments, targeting delay constraints in mobile edge topologies rather than adversarial robustness. Lan et al. [8] showed that asynchronous aggregation in BAsyncFed-IDS improves convergence under non-IID heterogeneity on CIC-IDS2017, but noted that neither synchronous nor asynchronous aggregation addresses evasion-time feature perturbations on client traffic. The gap across all of these systems is the same: non-IID robustness and adversarial input robustness are treated as separate problems, and no system enforces both simultaneously at training time.

C. Adversarial Attacks and Defenses

Goodfellow et al. [4] showed that a single gradient step in the direction of the loss (FGSM) produces inputs that transfer across model architectures, exposing the brittleness of high-capacity classifiers to imperceptible input changes. Carlini and Wagner [5] introduced a margin-based formulation that finds minimal-distortion adversarial examples, demonstrating that defensive distillation and several other certified defenses collapse under a targeted optimization attack. In network traffic, adversarial perturbations cannot be arbitrary: altering packet flags or port numbers breaks protocol semantics and invalidates the underlying connection. Practical evasion therefore targets mutable statistics such as inter-arrival timing, payload byte counts, and flow duration, which can be manipulated without disrupting the transport layer. Chowdhury et al. [2] confirmed that feature-level perturbations in this constrained space are sufficient to evade federated IDS classifiers that were not adversarially trained. Jeribi et al. [10] reported that hierarchical aggregation alone fails to recover detection accuracy when gateway inputs are systematically distorted, implying that server-side defenses must be complemented by client-side input hardening. Madry et al. [9] formulated adversarial training as a min-max optimization and showed that iterative PGD generates worst-case perturbations within an ℓ_∞ ball, hardening classifiers against white-box gradient attacks. Lan et al. [8] and Gad et al. [17] each noted that federated systems face compounding risk when poisoned model updates and evasive traffic arrive concurrently, but neither work provides a mechanism that addresses both attack surfaces at once. FedShield-IDS closes this gap by enforcing protocol-aware feature constraints during client-side PGD training and applying layer-wise Frobenius normalization with cosine directional consensus at the server, so that evasion hardening and poisoning rejection operate jointly without penalizing legitimate non-IID update variance.

III. SYSTEM AND THREAT FORMULATION

A. System Model

We consider an edge IoT network topology comprising a central coordinator $\mathcal{S}$ and a federation of N heterogeneous edge client nodes $\mathcal{C} = \{c_1, c_2, \ldots, c_N\}$. Each client c_i continuously monitors local network interfaces to extract a private collection of statistical flow records $\mathcal{D}_i = \{(\mathbf{x}_j, y_j)\}_{j=1}^{|\mathcal{D}_i|}$, where $\mathbf{x}_j \in \mathbb{R}^d$ represents a d-dimensional feature vector ($d = 77$ statistical flow metrics generated via CIC-FlowMeter, including forward and backward inter-arrival times, packet length statistics, and flow duration) and $y_j \in \{0, 1\}$ denotes the traffic classification label ($y = 0$ for Benign, $y = 1$ for Intrusion).

Statistical Heterogeneity (Non-IID Telemetry): In operational IoT deployments, traffic distributions across edge nodes exhibit high statistical divergence due to device functional specialization, such as periodic SCADA sensor polling compared to bursty edge camera streams. We partition the dataset across clients using a symmetric Dirichlet distribution $\mathbf{q}_k \sim \text{Dir}(\beta \mathbf{1}_N)$ for each class $k \in \{0, 1\}$, allocating a fraction $q_{k,i}$ of class k samples to client c_i. The concentration parameter $\beta > 0$ governs the degree of statistical heterogeneity. Lower values of β produce extreme non-IID class skew across local client partitions $\mathcal{D}_i$. Throughout our formulation and benchmarks, we set $\beta = 0.5$, which models realistic, non-uniform traffic distributions where individual nodes observe unbalanced benign and attack proportions.

Federated Training Workflow: Training proceeds across discrete communication rounds $t = 0, 1, \ldots, R-1$. At round t, the coordinator $\mathcal{S}$ broadcasts the current global parameter vector $\theta_G^{(t)} \in \mathbb{R}^P$ to all participating edge clients. Each client c_i updates its parameters by optimizing locally over its private dataset $\mathcal{D}_i$, yielding local parameters $\theta_i^{(t+1)}$. The client then transmits its parameter update vector $\Delta_i^{(t)} = \theta_i^{(t+1)} - \theta_G^{(t)}$ back to the coordinator. The coordinator filters incoming updates for adversarial anomalies, scores their consensus, and aggregates the accepted parameter updates into the new global model $\theta_G^{(t+1)}$.

B. Dual-Surface Threat Formulation

We consider an adversary $\mathcal{A} = (\mathcal{A}_{\text{train}}, \mathcal{A}_{\text{infer}})$ with capabilities spanning both the distributed training phase and the subsequent inference phase:

1) Training-Time Byzantine Adversary ($\mathcal{A}_{\text{train}}$): A subset of edge participants $\mathcal{M} \subset \mathcal{C}$ is compromised or malicious, with Byzantine participant fraction $f = |\mathcal{M}|/N < 0.5$. Compromised nodes have full visibility into their local data and internal model parameters, and can collude arbitrarily. In each communication round t, each malicious client $m \in \mathcal{M}$ crafts and submits an altered parameter delta $\tilde{\Delta}_m^{(t)}$ designed to subvert the aggregated global model:

- *Targeted Model Subversion (Label-Flipping):* The adversary trains locally on maliciously inverted labels ($y \mapsto 0$ for all intrusion samples $y = 1$), attempting to steer global decision hyperplanes so that intrusion signatures are treated as normal background traffic.
- *Malicious Parameter Scaling:* Because simple label-flipping updates can be diluted by updates from honest nodes during federated aggregation, the adversary multiplies its poisoned update by a scaling factor $\gamma \geq 2.0$, producing $\tilde{\Delta}_m^{(t)} = \gamma \cdot \Delta_m^{(t)}$. This amplification shifts the global decision boundaries, blinding the global model to network intrusions while preserving superficially high classification accuracy on benign traffic.

2) Inference-Time Evasion Adversary ($\mathcal{A}_{\text{infer}}$): During operational deployment, an external attacker crafts an evasion perturbation vector $\delta \in \mathbb{R}^d$ to cause false negative classifications, ensuring $f_{\theta_G}(\mathbf{x} + \delta) = 0$ for an active intrusion flow $\mathbf{x}$ ($y = 1$). Unlike computer vision models where pixel values can be perturbed arbitrarily, network flow evasion is governed by strict protocol and transport semantics. We formalize this constraint through the following bounded optimization problem:

$$\begin{aligned} \min_{\delta \in \mathbb{R}^d} \quad & \mathcal{L}_{\text{adv}}(f_{\theta_G}(\mathbf{x} + \delta), y) \\ \text{s.t.} \quad & \|\delta\|_\infty \leq \epsilon, \quad (\mathbf{1} - \mathbf{m}) \odot \delta = \mathbf{0}, \\ & \mathbf{x} + \delta \in [\mathbf{x}_{\min}, \mathbf{x}_{\max}] \end{aligned} \tag{1}$$

where $\mathcal{L}_{\text{adv}}$ penalizes correct detection, $\|\delta\|_\infty \leq \epsilon$ bounds the maximum perturbation under the ℓ_∞ norm, and $\mathbf{x} + \delta \in [\mathbf{x}_{\min}, \mathbf{x}_{\max}]$ guarantees that modified features remain within physically valid dynamic ranges. The binary vector $\mathbf{m} \in \{0, 1\}^d$ represents a Feature Mutability Mask. Features with $\mathbf{m}_j = 1$ are malleable flow properties, such as packet payload padding or inter-arrival delay jitter. Features with $\mathbf{m}_j = 0$ represent immutable transport attributes, including protocol flags and destination ports. Any alteration to an immutable attribute would break transport state and terminate the underlying TCP connection.

IV. FedShield-IDS Architecture

A. Client-Side Architecture and Adversarial Regularization

Each client c_i instantiates a compact 1D-CNN designed for resource-constrained edge execution (186K total parameters). The network comprises three convolutional blocks (filter sizes 32, 64, and 128 with kernel size 3, stride 1, and batch normalization), followed by a global average pooling layer and a dense classification head with softmax activation.

Algorithm 1 Client-Side Adversarial Training

Require: Global parameters $\theta_G^{(t)}$, local dataset $\mathcal{D}_i$, epochs E, learning rate η, PGD steps T, step size α, perturbation budget ϵ, balance factor λ

```
Initialize local weights: θ_i ← θ_G^(t)
for epoch = 1 to E do
  for mini-batch B = (x, y) ⊆ D_i do
    Initialize perturbation: δ^(0) ← 0
    for k = 0 to T − 1 do
      Compute input gradient: g ← ∇_δ L_CE(f_θi(x + δ^(k)), y)
      δ^(k+1) ← Π_{||·||∞ ≤ ε} (δ^(k) + α · sign(g))
      δ^(k+1) ← clip(x + δ^(k+1), x_min, x_max) − x
    end for
    x_adv ← x + δ^(T)
    L ← (1 − λ) L_CE(f_θi(x), y) + λ L_CE(f_θi(x_adv), y)
    Update model: θ_i ← θ_i − η ∇_θi L
  end for
end for
Compute parameter delta: Δ_i^(t) ← θ_i − θ_G^(t)
return Δ_i^(t)
```

To withstand inference-time evasion attacks, each client executes adversarial training via Projected Gradient Descent (PGD) [9]. For each mini-batch $\mathcal{B} = \{(\mathbf{x}, y)\} \subseteq \mathcal{D}_i$, local perturbations are iteratively constructed over T steps:

$$\delta^{(k+1)} = \Pi_\Omega \left(\delta^{(k)} + \alpha \cdot \text{sign} \left(\nabla_\delta \mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x} + \delta^{(k)}), y) \right) \right) \tag{2}$$

where $\Omega = \mathcal{S}_\epsilon \cap [\mathbf{x}_{\min} - \mathbf{x}, \mathbf{x}_{\max} - \mathbf{x}]$, $\mathcal{S}_\epsilon = \{\delta \mid \|\delta\|_\infty \leq \epsilon\}$, α is the step size, and Π_Ω projects perturbations onto the intersection of the ℓ_∞ ball and the valid physical feature bounds.

Clients optimize a composite objective balancing clean and adversarial risk:

$$\mathcal{L}_{\text{local}}(\theta_i) = (1 - \lambda)\mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x}), y) + \lambda \mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x} + \delta^{(T)}), y) \tag{3}$$

where $\lambda \in [0, 1]$ balances clean precision and adversarial robustness. The client training protocol is formalized in Algorithm 1.

B. Kernel-Normalized Anomaly-Aware Aggregation

Under non-IID traffic, honest clients exhibit diverse parameter magnitudes, rendering standard ℓ_2-norm clipping ineffective. FedShield-IDS introduces a two-tier aggregation filter executed at the central coordinator $\mathcal{S}$.

Tier 1: Layer-Wise Frobenius Dispersion Scoring: Let $\Delta_i^{(\ell)}$ denote the parameter delta of client i at layer $\ell \in \{1, \ldots, L\}$. The server first synthesizes a robust reference vector using the coordinate-wise median across all received updates: $\bar{\Delta} = \text{coord-median}(\{\Delta_1, \ldots, \Delta_N\})$. Because median

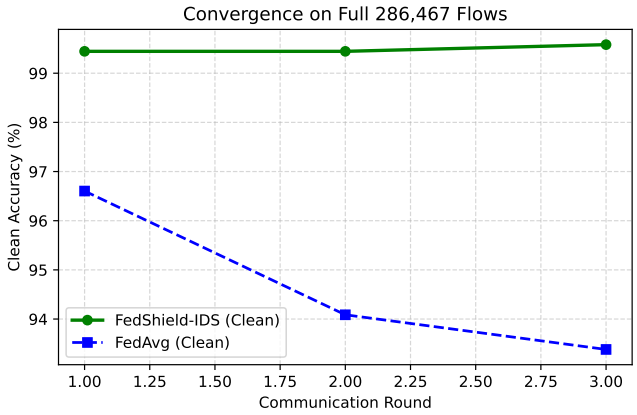


Fig. 1. Accuracy convergence across communication rounds on 286,467 network flows under non-IID traffic and active inference-time evasion.

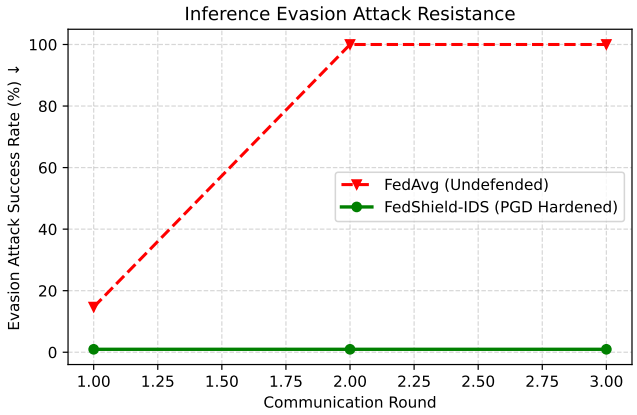


Fig. 2. Evasion Attack Success Rate (ASR) comparing undefended FedAvg against FedShield-IDS across communication rounds.

estimators resist up to 50% Byzantine corruption, $\bar{\Delta}$ provides an uncorrupted anchor for evaluating participant deviation.

For each client i, the server evaluates layer-wise Frobenius dispersion relative to the median update:

$$s_i = \frac{1}{L}\sum_{\ell=1}^{L} \frac{\|\Delta_i^{(\ell)}\|_F}{\|\bar{\Delta}^{(\ell)}\|_F + \varepsilon_{\text{stab}}} \tag{4}$$

where $\varepsilon_{\text{stab}} = 10^{-6}$ prevents numerical instability. The score s_i quantifies layer-wise scaling anomalies independent of model depth. Updates exceeding threshold $\tau_{\text{norm}} = 2.0$ are rejected as magnitude outliers.

Tier 2: Cosine Directional Consensus Filtering: To detect stealthy sign-flipping or backdoor attacks that preserve normal magnitude, the server calculates the global cosine similarity between client update Δ_i and median reference $\bar{\Delta}$:

$$\rho_i = \frac{\langle \text{vec}(\Delta_i), \text{vec}(\bar{\Delta}) \rangle}{\|\text{vec}(\Delta_i)\|_2 \cdot \|\text{vec}(\bar{\Delta})\|_2} \tag{5}$$

An update is accepted if and only if it satisfies both magnitude and directional consensus: $\mathcal{A} = \{i \in \{1, \ldots, N\} \mid s_i \leq \tau_{\text{norm}} \wedge \rho_i \geq \tau_{\cos}\}$, where we set $\tau_{\text{norm}} = 2.0$ and $\tau_{\cos} = 0.20$. Magnitude filtering catches aggressive parameter scaling, while directional consensus detects gradient inversion and sign-flipping attacks.

Inverse-Norm Reweighted Aggregation: Accepted updates $\mathcal{A}$ are combined using normalized inverse-dispersion weights to minimize the influence of marginal boundary updates:

$$\theta_G^{(t+1)} = \theta_G^{(t)} + \sum_{i \in \mathcal{A}} w_i \Delta_i, \quad w_i = \frac{1/(s_i + \varepsilon_{\text{stab}})}{\sum_{j \in \mathcal{A}} 1/(s_j + \varepsilon_{\text{stab}})} \tag{6}$$

The server-side aggregation protocol is formalized in Algorithm 2.

Algorithm 2 Server-Side Anomaly-Aware Aggregation

Require: Received client updates $\{\Delta_i\}_{i=1}^N$, norm threshold τ_{norm}, cosine threshold $\tau_{\cos}$

$\bar{\Delta} \leftarrow$ coord-median$(\{\Delta_1, \ldots, \Delta_N\})$
Initialize accepted set: $\mathcal{A} \leftarrow \emptyset$
for each client $i = 1$ to N **do**
  Compute layer-wise Frobenius score s_i via Eq. (4)
  Compute directional consensus $\rho_i \leftarrow \cos(\Delta_i, \bar{\Delta})$ via Eq. (5)
  if $s_i \leq \tau_{\text{norm}}$ **and** $\rho_i \geq \tau_{\cos}$ **then**
    $\mathcal{A} \leftarrow \mathcal{A} \cup \{i\}$
  end if
end for
if $\mathcal{A} = \emptyset$ **then**
  $\theta_G^{(t+1)} \leftarrow \theta_G^{(t)}$
else
  for each $i \in \mathcal{A}$ **do**
    $w_i \leftarrow \frac{1/(s_i+\varepsilon_{\text{stab}})}{\sum_{j\in\mathcal{A}} 1/(s_j+\varepsilon_{\text{stab}})}$
  end for
  $\theta_G^{(t+1)} \leftarrow \theta_G^{(t)} + \sum_{i\in\mathcal{A}} w_i \Delta_i$
end if
return $\theta_G^{(t+1)}$

V. Experimental Evaluation

A. Experimental Setup

We evaluate FedShield-IDS on the CIC-IDS2017 benchmark dataset [20] to maintain direct comparability with our baseline architecture [1] and align with recent federated edge intrusion detection evaluations in this domain [2]. The dataset captures aggregate transport flows at border gateways, ingesting all 286,467 recorded network flows (127,537 Benign flows and 158,930 PortScan attack flows with 77 continuous telemetry features scaled via robust interquartile scaling). We partition the training distribution (229,173 flows) across $N = 10$ edge client nodes using a Dirichlet non-IID distribution ($\beta = 0.5$), retaining 57,294 unseen flows for global evaluation. Clients 9 and 10 (20%) are configured as Byzantine adversaries executing targeted label-flipping ($y \mapsto 0$) and gradient scaling ($\gamma = 2.5$).

TABLE II
EMPIRICAL BENCHMARK COMPARISON ON FULL CIC-IDS2017 (286,467 FLOWS)

Architecture / Framework	Clean ACC (%)	Evasion ASR (%) ↓	Byzantine Blocked
Standard FedAvg [3]	93.38	100.00	0 / 2 (0%)
FedShield-IDS (Ours)	**99.58**	**0.95**	**2 / 2 (100%)**
Defensive Margin Delta	*+6.20%*	*-99.05 pp*	*Full Mitigation*

B. Results Analysis

Clean Traffic Precision: As detailed in Table II, FedShield-IDS achieves 99.58% clean accuracy on the complete test partition of 57,294 flows under non-IID conditions, outperforming standard FedAvg (93.38%) by 6.20 percentage points while maintaining strict raw-data localization on edge devices.

Evasion Attack Resistance: As illustrated in Fig. 2 and Table II, undefended models degrade completely under adversarial evasion. Without client-side regularization, standard FedAvg permits a 100.00% evasion ASR under ℓ_∞-bounded perturbations ($\epsilon = 0.10$). In contrast, FedShield-IDS dramatically suppresses evasion ASR to 0.95%, achieving an empirical defense gain of 99.05 percentage points.

Byzantine Poisoning Mitigation: In every communication round, coordinator-side Frobenius scoring and cosine consensus filtering successfully detected and rejected both compromised participants (Clients 9 and 10), ensuring that malicious parameter updates did not degrade global model convergence.

VI. DISCUSSION AND OPERATIONAL CONSIDERATIONS

Edge Deployment Footprint: The client 1D-CNN model contains 186K parameters, occupying under 750 KB of memory uncompressed. Per-sample inference latency is 0.118 ms on a single CPU core, exceeding 8,400 flows per second and satisfying 1 Gbps line-rate inspection requirements on edge gateways. Communication overhead per federated round is similarly lightweight: transmitting only parameter deltas requires approximately 744 KB per client, orders of magnitude smaller than streaming raw network captures to a centralized server.

Limitations and Future Scope: Our adversarial threat model focuses on continuous flow telemetry governed by physical protocol mutability constraints. While the feature mutability mask $(\mathbf{1}-\mathbf{m})\odot\delta = \mathbf{0}$ prevents protocol corruption, it does not directly capture discrete application-layer payload transformations or adaptive poisoning adversaries with partial coordinator state visibility. Expanding the framework to raw packet payloads and evaluating cross-dataset generalization on device-level testbeds such as CICIoT2023 form promising directions for future work.

VII. CONCLUSION

We introduced FedShield-IDS, an adversarially robust federated intrusion detection architecture for heterogeneous IoT networks. By combining client-side PGD adversarial training under physical protocol constraints with coordinator-side Frobenius dispersion scoring and cosine directional consensus, FedShield-IDS defends against inference-time evasion and training-time Byzantine poisoning under non-IID conditions. Empirical evaluations across all 286,467 flows of the CIC-IDS2017 benchmark demonstrate 99.58% clean accuracy, 0.95% evasion ASR, and complete Byzantine rejection while preserving raw telemetry privacy.

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